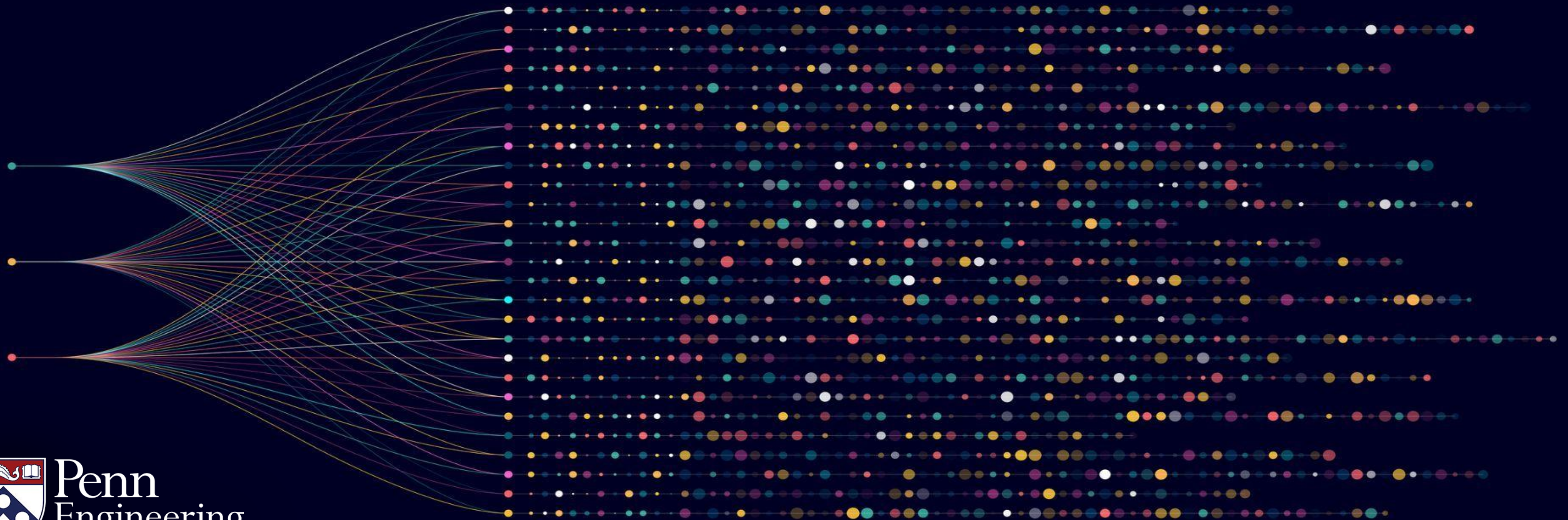


Algorithms for Big Data

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Continuous random variables



Continuous random variables

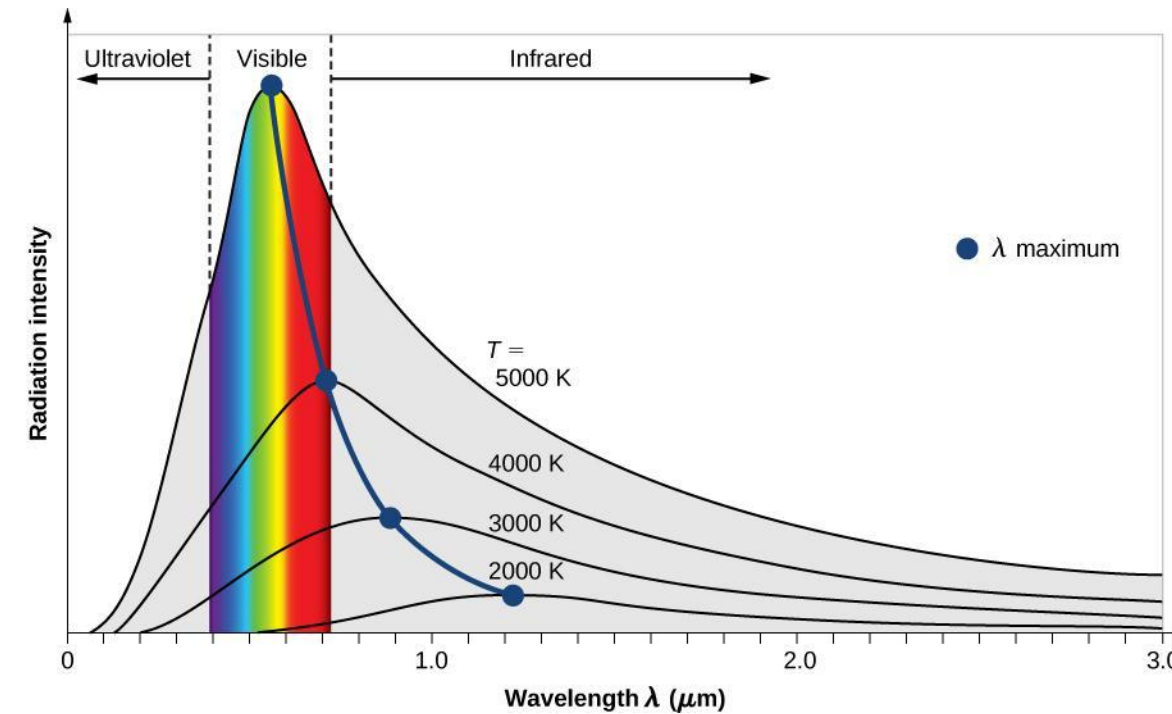
So far, our random variables have been discrete

Most of the course: suffices to think about discrete random variables

However, some parts of this course: necessary to study continuous random variables

Continuous random variables: where do they arise?

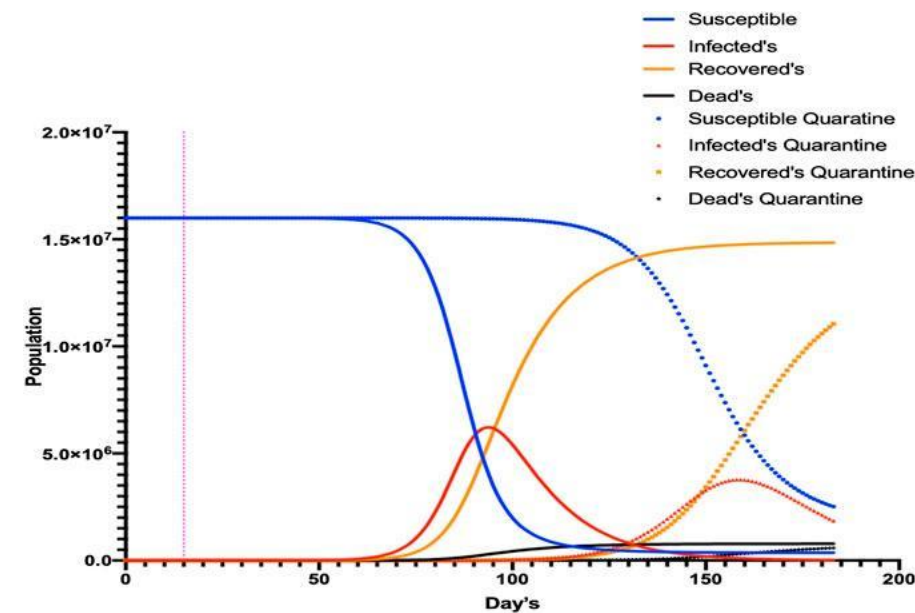
In theory and practice, many random variables are modeled as continuous random variables



Example #1: Planck's formula for blackbody radiation

Continuous random variables: where do they arise?

In theory and practice, many random variables are modeled as continuous random variables



Example #2: Models for spread of infectious diseases in epidemiology

Probability density function

Discrete random variables $\mathbf{X}$ are given by $\Pr[\mathbf{X} = z]$ for all points z in the support of $\mathbf{X}$
(support of $\mathbf{X}$ is the set of points z such that $\Pr[\mathbf{X} = z] \neq 0$)

For continuous random variables $\mathbf{X}$, meaningless to specify $\Pr[\mathbf{X} = z]$ -- if $\mathbf{X}$ is continuous, the probability that it takes any specific value x is zero

Instead, we specify a continuous random variable using a probability density function $p(\cdot)$
(abbreviated as pdf)

Probability density function

For a continuous random variable $\mathbf{X}$, the pdf $p(\cdot)$ tells us the “density” of $\mathbf{X}$ at the point

For small δz , $\Pr[\mathbf{X} \in [z, z + \delta z]] \sim p(z) \cdot \delta z$.

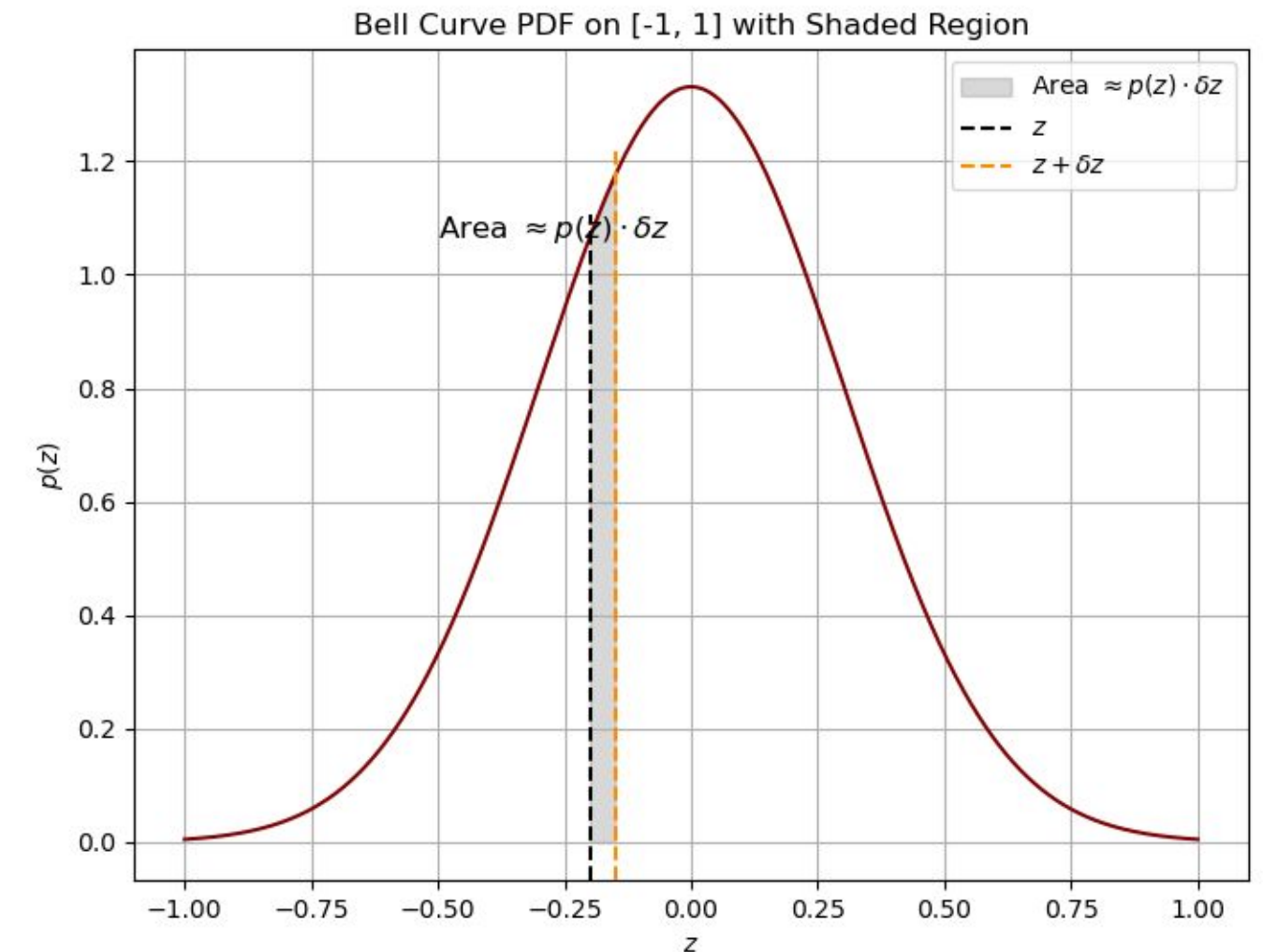
Or in the calculus sense,

$$\lim_{\delta z \rightarrow 0} \Pr[\mathbf{X} \in [z, z + \delta z]] = p(z) \cdot \delta z$$

If $p(\cdot)$ is given as a graph, then

$\Pr[\mathbf{X} \in [z, z + \delta z]] = \text{area of gray shaded region}$

As $\delta z \rightarrow 0$, area of gray region = $p(z) \cdot \delta z$.



Probability density function

So far, we have considered $\Pr[\mathbf{X} \in [z, z + \delta z]]$, i.e., width of the interval $\delta z \rightarrow 0$

What happens in the general case? How do we get $\Pr[\mathbf{X} \in [z_1, z_2]]$ for general z_1, z_2 ?

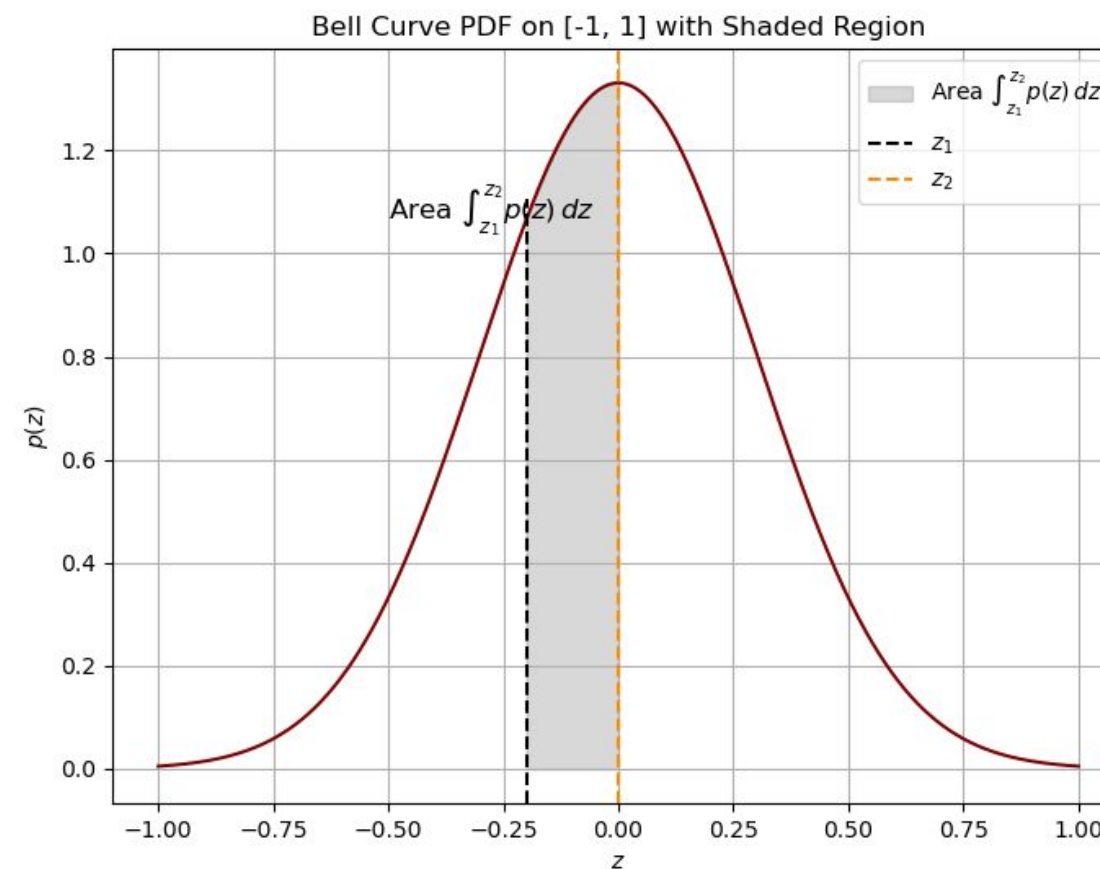
Obvious idea: break $[z_1, z_2]$ into many small intervals and add up the probabilities

In the language of calculus, $\Pr[\mathbf{X} \in [z_1, z_2]] = \int_{z_1}^{z_2} p(z)dz.$ — (1)

Probability density function

Thus, if the pdf of $\mathbf{X}$ is $p(\cdot)$,

$$\Pr[\mathbf{X} \in [z_1, z_2]] = \int_{z_1}^{z_2} p(z) dz. \quad - (1)$$



Dictionary between “discrete” and “continuous” worlds

For a continuous random variable $\mathbf{X}$

$$\Pr[\mathbf{X} \in [z_1, z_2]] = \int_{z_1}^{z_2} p(z) dz.$$

Compare it with the discrete world – For a random variable $\mathbf{X}$,

$$\Pr[\mathbf{X} \in [z_1, z_2]] = \sum_{z_1 \leq z \leq z_2} \Pr[\mathbf{X} = z].$$

Dictionary: 1. “Sum” gets replaced by an “integral”

2. $\Pr[\mathbf{X} = z]$ gets replaced by $p(z)$

Dictionary between “discrete” and “continuous” worlds

Discrete r. v.

Continuous r. v.

Law of total probability:

$$\sum_z \Pr[X = z] = 1$$

Law of total probability:

$$\int p(z) dz = 1$$

Integral should go from $z = -\infty$ to ∞ .

Can exclude sets where $p(z) = 0$.

Expectation: $E[X] := \sum_z z \cdot \Pr[X = z]$

Expectation: $E[X] := \int z \cdot p(z) dz$

Second moment: $E[X^2] = \sum_z z^2 \cdot \Pr[X = z]$

Second moment: $E[X^2] = \int z^2 \cdot p(z) dz$

Dictionary between “discrete” and “continuous” worlds

Discrete r. v.

Continuous r. v.

More generally,

$$E[f(\mathbf{X})] = \sum_z f(z) \cdot \Pr[\mathbf{X} = z].$$

More generally,

$$E[f(\mathbf{X})] = \int f(z) \cdot p(z) dz.$$

Cumulative distribution function

Cumulative distribution function (cdf) for a continuous random variable $\mathbf{X}$:

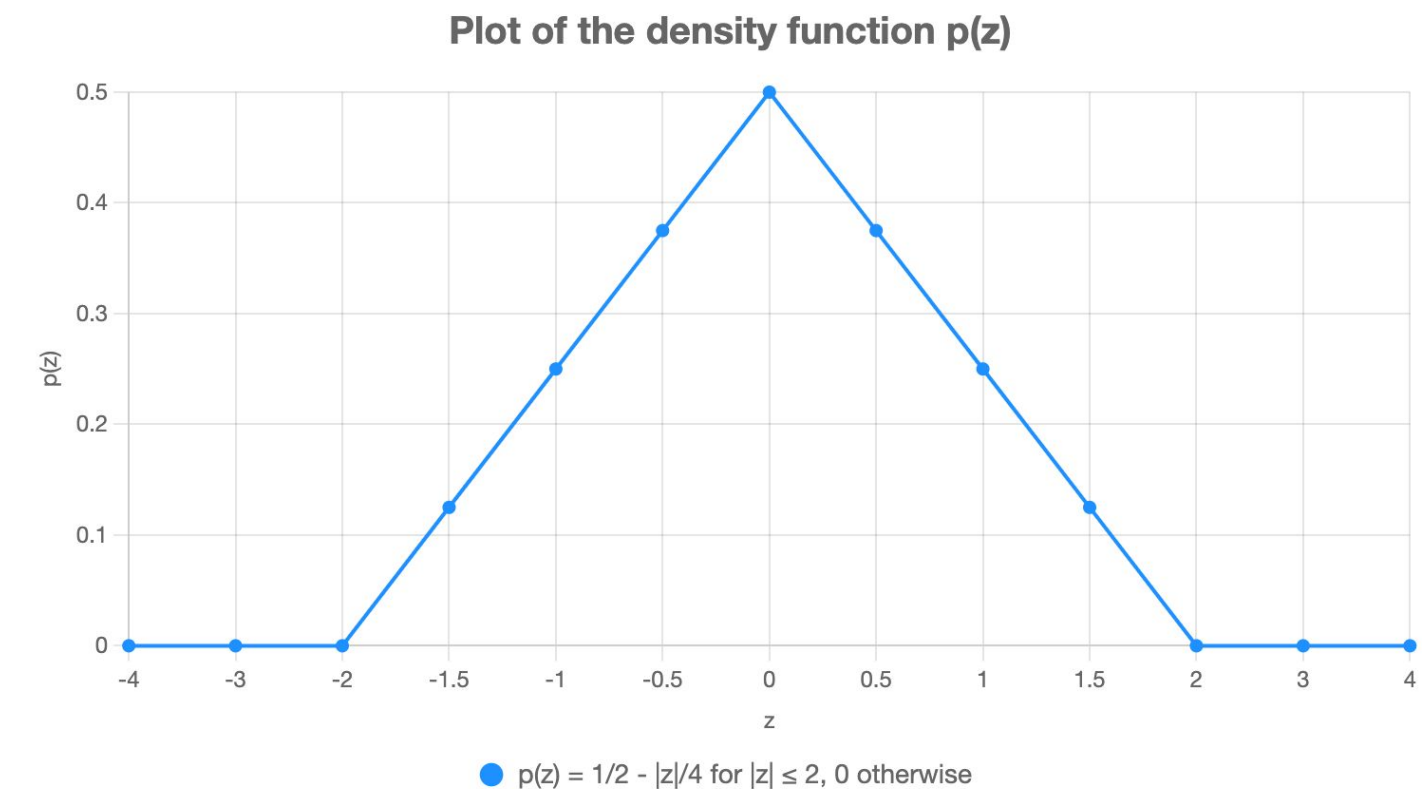
$$\varphi(t) := \Pr[\mathbf{X} \leq t].$$

Suppose the density function is given by $p(z)$. Then,

$$\text{it follows: } \varphi(t) = \int_{z=-\infty}^t p(z) dz.$$

Example: Density function of r.v. $\mathbf{X}$ given by $p(z)$

$$p(z) = \begin{cases} \frac{1}{2} - \frac{|z|}{4} & \text{if } |z| \leq 2 \\ 0 & \text{if } |z| > 2 \end{cases}$$



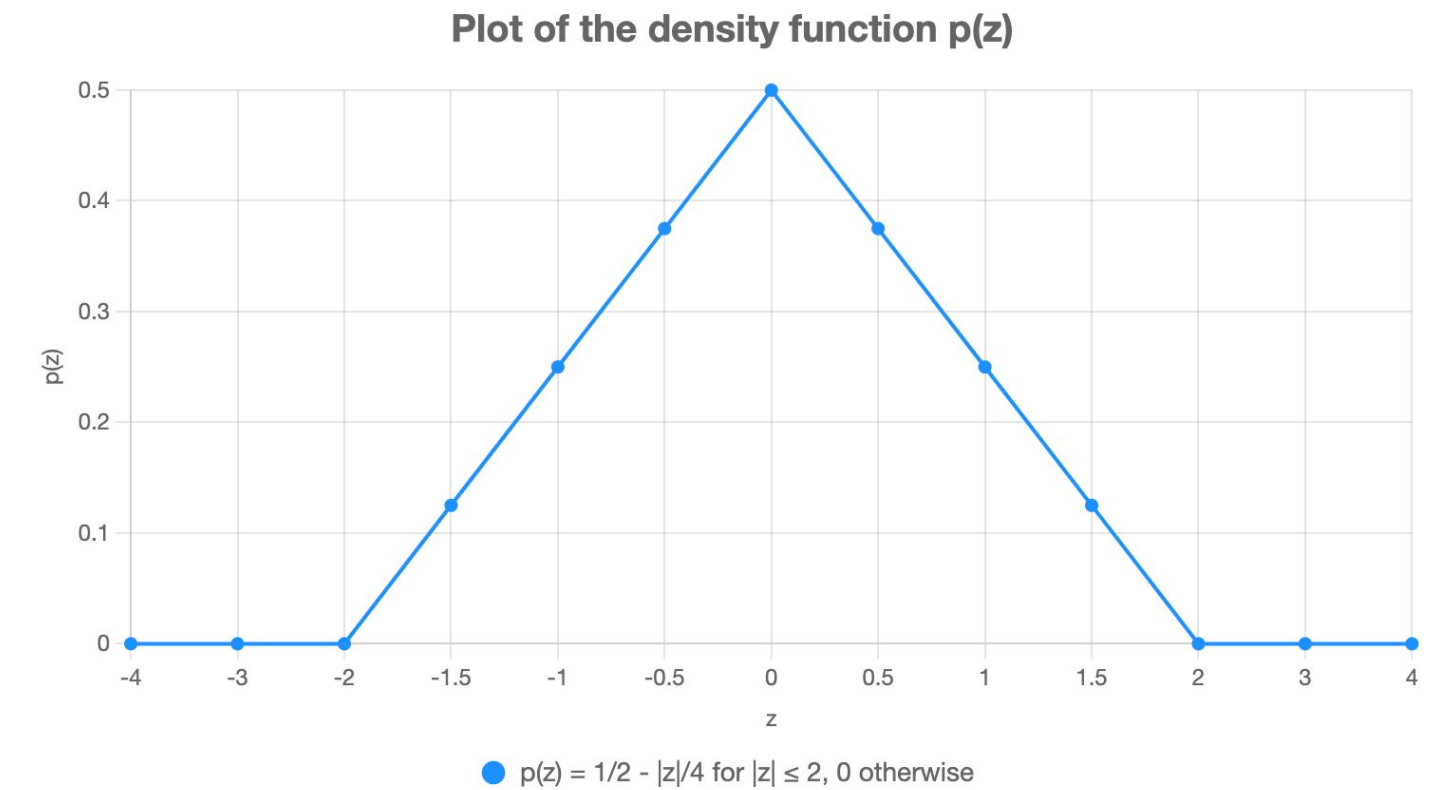
Cumulative distribution function

$$\varphi(t) = \Pr[X \leq t] = \int_{z=-\infty}^t p(z) dz.$$

1. If $t \leq -2$, $\varphi(t) = 0$ (because $p(z) = 0$ if $z \leq -2$)

2. If $-2 \leq t \leq 0$, $\varphi(t) = \int_{z=-\infty}^t p(z) dz = \int_{z=-2}^t p(z) dz$

$$= \int_{z=-2}^t \left(\frac{1}{2} - \frac{|z|}{4} \right) dz = \int_{z=-2}^t \left(\frac{1}{2} + \frac{z}{4} \right) dz = \frac{1}{2}t + \frac{t^2}{8}.$$



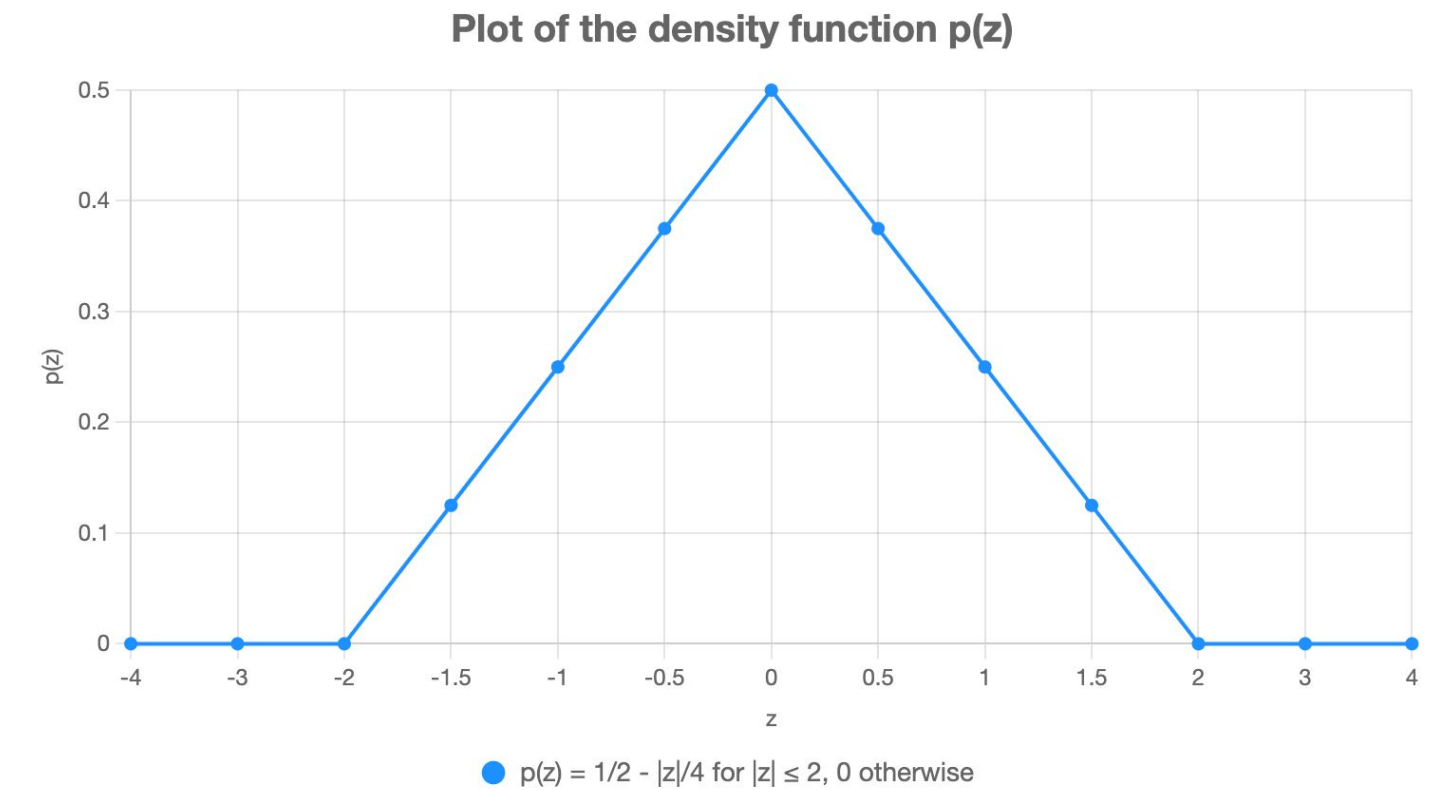
Cumulative distribution function

$$\varphi(t) = \Pr[\mathbf{X} \leq t] = \int_{z=-\infty}^t p(z) dz.$$

3. If $0 \leq t \leq 2$, $\varphi(t) = \int_{z=-\infty}^t p(z) dz$

$$= \int_{z=-\infty}^0 p(z) dz + \int_{z=0}^t p(z) dz = \frac{1}{2} + \frac{t}{2} - \frac{t^2}{8}.$$

4. Finally, for $t \geq 2$, $\varphi(t) = \Pr[\mathbf{X} \leq t] = 1$
(pdf vanishes for any $z \geq 2$).



Relation between cdf and pdf

In summary, if the pdf of a random variable $\mathbf{X}$ is $p(\cdot)$ and the cdf is $\varphi(\cdot)$, then

$$\varphi(t) = \int_{z=-\infty}^t p(z) dz.$$

Thus, if $\varphi(\cdot)$ is differentiable, then $p(z) = \frac{d\varphi(z)}{dz}$, i.e., $p(\cdot)$ is the derivative of $\varphi(\cdot)$.

In many application, $\varphi(\cdot)$ is differentiable and thus, we can easily get the density function $p(\cdot)$ in terms of $\varphi(\cdot)$.

Continuous random variables: summary

- Continuous random variables occur at many places – theory and practice
- Specified by the probability density function pdf
- Dictionary between discrete and continuous worlds – summation replaced by integrals and point probabilities replaced by pdf